

Rajakumar Arul

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in [rajakumar-arul](#)

Associate Professor 🇮🇳 Nationality: India | 🎂 Date of birth [REDACTED] ❤️ Marital Status: Married

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Profile Summary

- Result-oriented academic and researcher with 8+ years of experience in Research & Development, curriculum design, and industry collaboration.
- Experienced in research proposal writing, project management using Agile (SCRUM, Kanban), and supervision of postgraduate projects.
- Recipient of Anna Centenary Fellowship for full-time Ph.D.; authored 15+ international publications, including first-author articles in IEEE Transactions on Industrial Informatics.

Research Interests

- Cyber Security and Cryptography
- Quantum Computing
- Post-Quantum Cryptography
- Wireless Technology & 5G Networks
- Machine Learning and IoT

Experience

- Jun 2021 – **Assistant Professor / Associate Professor**, Vellore Institute of Technology (VIT) Chennai.
Present
 - **HoD | B.Tech CSE (Quantum Computing), Mar 2026 – Present.**
 - Teaching, research leadership, grant management, and mentoring postgraduate theses.
 - Using Agile methodology for project development; establishing research collaborations across national and international labs.
 - Mentor, iSpace Club (VIT Chennai), Jun 2023 – Present.
- Jul 2018 – **Assistant Professor**, Amrita School of Engineering, Bengaluru.
- May 2021
 - Mentored Face Technical Club; taught core CS courses and guided projects in IoT Security and Blockchain.
 - Received Chancellor's Publication Award at Amrita Innovation and Research Awards (2020) for publication in IEEE TII.
- Mar 2018 – **Senior Researcher (Part-Time, Remote)**, RNS Solutions, Singapore.
- Nov 2018
 - Contributed to applied research in blockchain and cybersecurity use-cases.
- Jul 2013 – **Full-Time Research Scholar**, Anna University, Chennai.
- May 2018
 - Conducted doctoral research in Information and Communication Engineering.

Teaching (Selected)

Data Structures & Algorithms; Software Engineering; Cloud Computing;
Quantum Algorithms; Post-Quantum Cryptography; DevSecOps

Supervision & Mentoring

Mentor for the Winning Team of the Researcher Discovery Platform Competition, Techfest IIT Bombay (2021).

Supervised postgraduate student projects on IoT Security and Blockchain.

Education

- 2013 – 2018 **Ph.D. in Information and Communication Engineering**, *Madras Institute of Technology, Anna University, Chennai*, (Full-Time).
Fellowship: Anna Centenary Research Fellowship
- 2011 – 2013 **M.E. in Computer Science and Engineering**, *Madras Institute of Technology, Anna University, Chennai*.
- 2007 – 2011 **B.E. in Computer Science and Engineering**, *Vidyaa Vikas College of Engineering & Technology, Tiruchengode*.
- 2021 **MBA (Human Resource Management) (Part-Time)**, *Annamalai University, Chidambaram*.

Certifications

- ★ Certified in OffSec CyberCore (OSCC), Offensive Security.
- ★ Qualified Qniverse Developer Certification, CDAC - India.
- ★ Certified Scrum Master (CSM), Scrum Alliance.
- ★ Certified Apple Teacher (Apple Applications & SWIFT).
- ★ Oracle Java Foundations Certified Associate.

Technical Skills

Operating Systems	macOS, Linux, Windows
Languages & Platforms	C, Java, Swift, AWS, GCP, Ethereum, Go
Tools	Xcode (iOS/macOS), JIRA, ns-3

Awards

- Anna Centenary Fellowship for full-time Ph.D., Anna University, Chennai.
- 2020 Chancellor's Publication Award, Amrita Innovation and Research Awards, for publication in IEEE Transactions on Industrial Informatics.
- 2022, 2023, 2024 Faculty Research Award for Project & Publication, VIT Chennai
- 2026 Service Award for Completing 5 Years of Service, VIT Chennai

Grants & Project Experience

- PI Side-channel attack exploration in Smart Wearable Devices; SERB-India; INR 18.30 Lakhs; 06 Oct 2022 – 30 Oct 2025.
- Co-PI MeitY Quantum Computing Applications Lab in association with AWS; MeitY India & AWS; USD 8,900; 10 Apr 2023 – 10 Apr 2025.
- PI Design of a post-quantum cryptographic algorithm (lattice-based) for quantum-threat mitigation; Mitacs, Canada; USD 9,000; 14 May 2024 – 31 May 2025.

Journal Publications

- 2026 *Shaik Jamaludeen, Amrit Pal, Rajakumar Arul, and Abishi Chowdhury. "AB-NAECP: Attention Based Neural Architecture for Electricity Consumption Prediction." IEEE Access (2026).*

- 2025 *Columbus Chinnappan*, Christopher, Palani Thanaraj Krishnan, Elakiya Elamaran, **Rajakumar Arul**, and T. Sunil Kumar. “Quantum Computing: Foundations, Architecture and Applications.” *Engineering Reports* **7** (2025).
- 2024 BA, Mohan, Prakash G. L., **Rajakumar Arul**, and Kalaipriyan Thirugnanasambandam. “Machine learning-driven strategies for customer retention and financial improvement.” *Archives for Technical Sciences* **2**, no. 31 (2024): 269–283.
- 2022 *Radha Doraisamy*, Minal Moharir, and **Rajakumar Arul**. “Congestion-aware and game-based odd-even adaptive routing in network-on-chip many-core architecture.” *Indonesian Journal of Electrical Engineering and Computer Science* **28**, 962–972 (2022).
- T. P. Latchoumi*, K. Raja, Y. Jyothi, K. Balamurugan, and **Rajakumar Arul**. “Mine safety and risk prediction mechanism through nanocomposite and heuristic optimization algorithm.” *Measurement: Sensors* **23**, 100390 (2022).
- Roobaea Alroobaea*, **Rajakumar Arul**, Saeed Rubaiee, Fahd S. Alharithi, Usman Tariq, and Xincan Fan. “AI-assisted bio-inspired algorithm for secure IoT communication networks.” *Cluster Computing* **25**, 1805–1816 (2022).
- 2021 **Rajakumar Arul**, *Roobaea Alroobaea*, Seifeddine Mechti, Saeed Rubaiee, Murad Andejany, Usman Tariq, and Saman Iftikhar. “Intelligent data analytics in energy optimization for the Internet of Underwater Things.” *Soft Computing* **25**, 12507–12519 (2021).
- Rajakumar Arul**, *Yasser D. Al-Otaibi*, Waleed S. Alnumay, Usman Tariq, Umar Shoaib, and M. D. Jalil Piran. “Multi-modal secure healthcare data dissemination framework using blockchain in IoMT.” *Personal and Ubiquitous Computing*, 1–13 (2021).
- Rajakumar Arul**, *Roobaea Alroobaea*, Usman Tariq, Ahmed H. Almulihi, Fahd S. Alharithi, and Umar Shoaib. “IoT-enabled healthcare systems using blockchain-dependent adaptable services.” *Personal and Ubiquitous Computing*, 1–15 (2021).
- 2019 **Rajakumar Arul**, *Gunasekaran Raja*, Alaa Omran Almagrabi, Mohammed Saeed Alkatheiri, Sajjad Hussain Chauhdary, and Ali Kashif Bashir. “A quantum-safe key hierarchy and dynamic security association for LTE/SAE in 5G scenario.” *IEEE Transactions on Industrial Informatics* **16**, 681–690 (2019).
- Ali Kashif Bashir*, **Rajakumar Arul**, Shakila Basheer, Gunasekaran Raja, Ramkumar Jayaraman, and Nawab Muhammad Faseeh Qureshi. “An optimal multitier resource allocation of cloud RAN in 5G using machine learning.” *Transactions on Emerging Telecommunications Technologies* **30**, e3627 (2019).
- Muhammad Shakil*, Alaelddin Fuad Yousif Mohammed, **Rajakumar Arul**, Ali Kashif Bashir, and Jun Kyun Choi. “A novel dynamic framework to detect DDoS in SDN using metaheuristic clustering.” *Transactions on Emerging Telecommunications Technologies*, e3622 (2019).
- 2018 **Rajakumar Arul**, *Gunasekaran Raja*, Ali Kashif Bashir, Junaid Chaudry, and Amjad Ali. “A console GRID leveraged authentication and key agreement mechanism for LTE/SAE.” *IEEE Transactions on Industrial Informatics* **14**, 2677–2689 (2018).
- 2017 *Ramkumar Jayaraman*, Gunasekaran Raja, Dipak Ghosal, **Rajakumar Arul**, and Sabareesh Kumar A. “A compatibility vector technique for cooperative scheduling and channel assignment algorithm in broadband wireless networks.” *Mobile Networks and Applications* **22**, 730–742 (2017).
- 2016 **Rajakumar Arul**, *Gunasekaran Raja*, Kottilingam Kottursamy, Pavithra Sathiyarayanan, and Swaminathan Venkatraman. “User path prediction-based key caching and authentication mechanism for broadband wireless networks.” *Wireless Personal Communications* **94**, 2645–2664 (2016).

Conference Publications

- 2026 N. Rakesh, B. A. Mohan, U. Kumaran, G. L. Prakash, Mohamed Shakir, and **Rajakumar Arul**. “Machine Learning Approaches for Early Diagnosis of Neurodegenerative Conditions.” In *The First International Conference on Recent Trends in Artificial Intelligence, Cyber Security, and Embedded Systems (ICRTACES 2024)*, Tiruchirappalli, India, 19–20 September 2024. Published in *AIP Conference Proceedings*, Vol. 3345, No. 1, Article 020149, 07 January 2026.
- 2025 Divyaansh Agarwal, **Rajakumar Arul**, and Kalaipriyan Thirugnanasambandam. “Review on the Implications of CRYSTALS-Kyber in LTE Networks.” In *4th International Conference on Security & Privacy (ICSP 2025)*, Department of Mathematics, National Institute of Technology Rourkela, India, 05–07 December 2025.
- 2024 Shakil Muhammad, Adnan Mujahid Khan, Azka Qureshi, **Rajakumar Arul**, and Kalaipriyan Thirugnanasambandam. “Advanced Automated System for Optimal Management in Public Health Emergencies.” In *International Conference on Computing and Communication Networks (ICCCN)*, pp. 629–638, 2024-10-17.
- 2023 **Rajakumar Arul**, B. Sivaselvam, and Ankush Choudhary. “A Novel Key Generation Algorithm Utilizing Lattice-Based Cryptography Principles.” In *Third International Conference on Computing and Communication Networks (ICCCN 2023)*, Manchester Metropolitan University, UK.
- 2022 “Prediction of Dementia using Whale Optimisation Algorithm based CNN.” In *International Conference on Machine Intelligence and Signal Processing (MISP 2022)*, NIT Raipur, India. *Accepted*.
- 2021 Auxilia Michael, K. Raja, Kannan Kaliyan, and **Rajakumar Arul**. “A Novel Secure Data Processing Mechanism in IoT Using Deep Learning with Ontology.” In *International Conference on Computing and Communication Networks (ICCCN 2021)*, Manchester Metropolitan University, UK.
- 2020 D. Samal and **Rajakumar Arul**. “A Novel Privacy Preservation Scheme for Internet of Things Using Blockchain Strategy.” In *International Conference on Communication, Computing and Electronics Systems (ICCES)*, (2020).
- Rajakumar Arul**, K. Vishnu, A. Eleyan, and A. K. Bashir. “The Authenticity of Information on Social Media.” *IEEE* (venue details not specified) (2020).
- 2019 K. Kannan, K. Raja, **Rajakumar Arul**, P. K. Nizar Banu. “E-Business Decision Support System for Online Shopping using MAS with Ontology and JADE Methodology.” *International Research Conference of Uva Wellassa University (IRCWWU 2019)*, Poster, p. 88 (Feb 7–9, 2019).
- 2017 R. J. Rajakumar Arul and Gunasekaran Raja. “Simple & Secure Mechanism for Establishing Connection between D2D Communication in 5G Scenario.” *International Journal of Engineering Research in Computer Science and Engineering* **4**(8), 173–178 (2017).
- 2016 G. Raja, R. Jayaraman, R. Arul, and K. Kottursamy. “Network Coding with Buffer Scheme in Multicast for Broadband Wireless Network.” *International Journal of Computer, Electrical, Automation, Control and Information Engineering* **10**(6), 898–903 (2016).
- R. Jayaraman, G. Raja, **Rajakumar Arul**, S. B. M. Baskaran, A. Arulappan, and K. Kottursamy. “Fault-tolerant Interworking Authentication Model for Non 3GPP – 3GPP Access.” In *International Conference on Innovations in Engineering and Technology (ICET 2016)*, Singapore, pp. 75–82 (May 21–22, 2016).
- R. Jayaraman, G. Raja, **Rajakumar Arul**, S. B. M. Baskaran, A. Arulappan, and K. Kottursamy. “BAY-RACE: Bayesian-Resource Allocation Conditional Entities Algorithm under Uncertainty in Distributed WMN.” In *International Conference on Innovations in Engineering and Technology (ICET 2016)*, Singapore, pp. 66–74 (May 21–22, 2016).

A. S. Arunkumar Arulappan, G. Raja, S. B. M. Baskaran, **Rajakumar Arul**, Kottilingam Kottursamy, R. Jayaraman. “Sensor-to-Vehicle Communication by using VANETs for Urban Parking.” In *International Conference on Innovations in Engineering and Technology (ICET 2016)*, Singapore, pp. 19–27 (2016).

- 2014 R. Gunasekaran, **Rajakumar Arul**, K. Raja, J. Ramkumar, S. K. S. Kumar. “An efficient handover prediction & initiation algorithm for vehicular communication in 4G wireless networks.” *Lecture Notes on Information Theory* **2**(2), 121–128 (June 2014).

R. Sheeba, S. Ramkumar J., Gunasekaran R., and **Rajakumar Arul**. “A Survey on Scheduling Mechanisms in Broadband Wireless Networks.” *Lecture Notes on Information Theory* **2**(3), 214–219 (September 2014).

Book Chapters

- 2023 Arul, R., K. Thirugnanasambandam, and M. Kiruthigga. “Internet of Things and medical imaging AI systems.” In: *Medical Imaging Informatics*, Chapter 10, pp. 205–220.
- 2021 Arul, R., Kothandaraman, R., Kannan, N.B. “Blockchain for Industry 4.0: An Assessment of Blockchain Adoptability.” In: *Industry 4.0 Interoperability, Analytics, Security, and Case Studies*, CRC Press.
- 2021 Arul, R., Basheer, S., Abbas, A., Bashir, A.K. “Role of Deep Learning Algorithms in Securing Internet of Things Applications.” In: *Deep Learning for Internet of Things Infrastructure*, CRC Press.
- 2019 Arul, R. “Ensemble learning mechanisms for threat detection.” In: *Machine Learning and Cognitive Science Applications in Cyber Security*, IGI Global.

Editorial Experience

- 2023 – Editorial Board Member, *IEEE Future Directions Newsletter: Technology, Policy and Ethics*.
Present
- 2022 – Editorial Board Member, *Research Reports on Computer Science*, Universal Wiser Publisher.
Present

Professional Memberships

- ⊙ Senior Member, IEEE
- ⊙ Professional Member, ACM
- ⊙ Member, Internet Society

Technical/Community Contributions

- ▷ Local Coordinator, Global Initiative of Academic Networks (GIAN), Dec 2024 – Present.
- ▷ Mentor – iSpace Club (VIT Chennai), Jun 2023 – Present.
- ▷ Mentor – Face Technical Club (Amrita School of Engineering), Dec 2019 – May 2021.

Corporate Training

- ★ Walmart Global Tech, Bengaluru, India
- ★ Thalir Capitals Pvt. Ltd., Mannargudi, India